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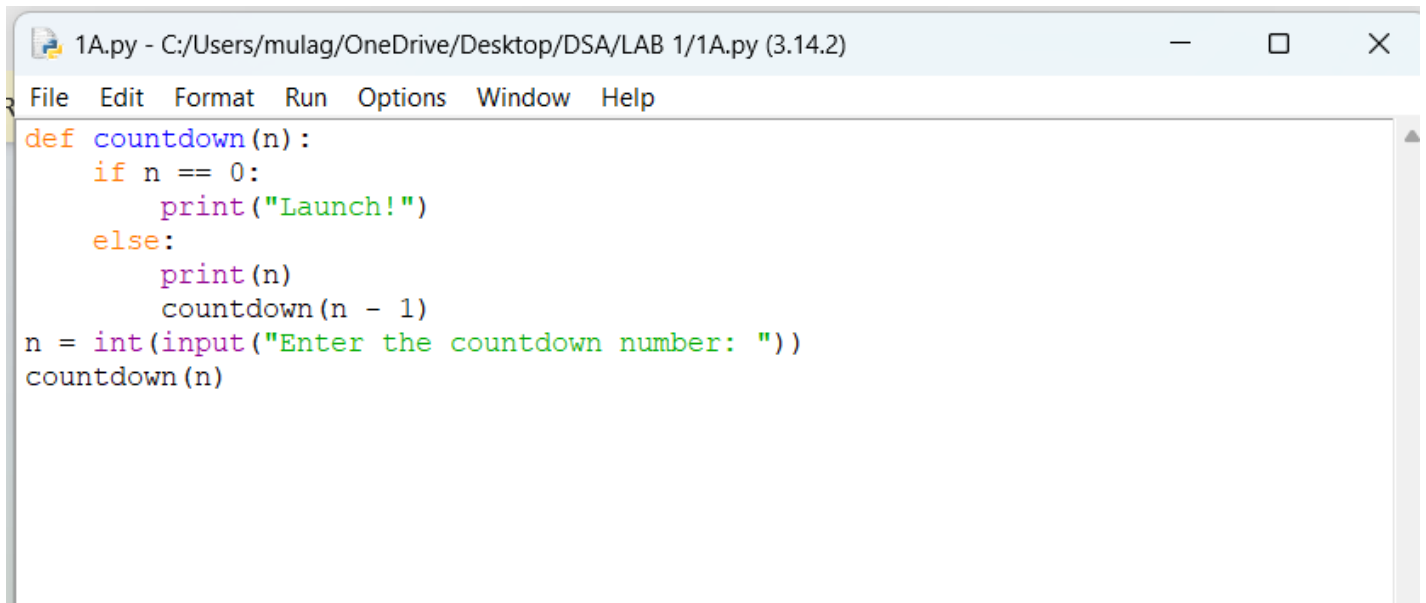
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1	<u>Lab-1:</u> 1.A rocket launch system displays a countdown before launch. Instead of using a loop implement the countdown using recursion .write a recursion python program to display a countdown from n to 1 followed by “launch”. 2.Banks calculate the growth of an investment using compound interest the formula requires calculating powers. Which can be implemented recursively. Write a recursive program to calculate p^n(power),where p=principal growth factor, n= number of years 3. An HR department stores employee ID’s in a list. Develop a recursive program to search for a given employee ID .write a recursive function to search for an employee ID in a list. 4. In a logistics company the number of possible ways to arrange n different parcels is calculated using the factorial formula(n!). develop a recursive program to calculate the factorial of a given number. 5. write a recursive python program to print the first n terms of the Fibonacci series.	3-10	
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Lab-1

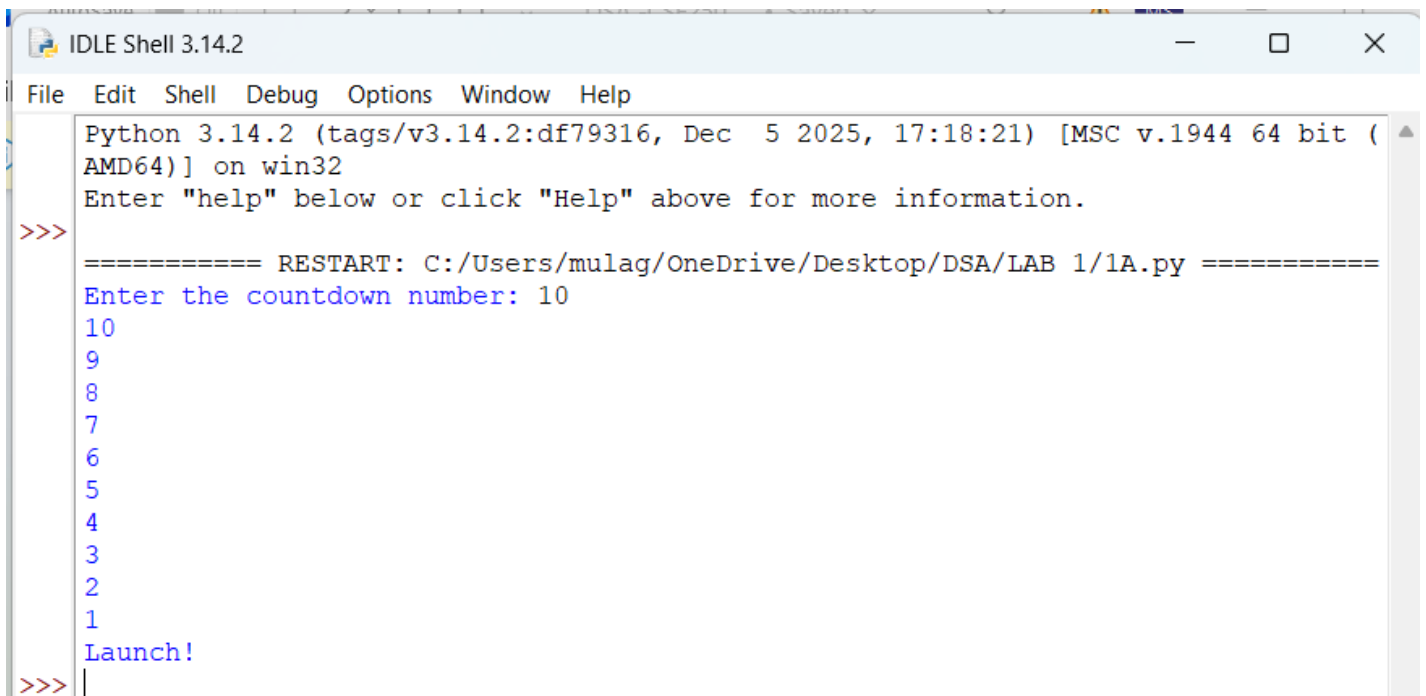
1)AIM: A rocket launch system displays a countdown before launch. Instead of using a loop implement the countdown using recursion .write a recursion python program to display a countdown from n to 1 followed by “launch”.

PROGRAM:

A screenshot of a Python IDE window titled '1A.py - C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1A.py (3.14.2)'. The window contains a Python program that defines a recursive function 'countdown(n)'. The function prints 'Launch!' if n is 0, and otherwise prints n and calls itself with n-1. The program prompts the user to 'Enter the countdown number:' and then calls the function. The code is as follows:

```
def countdown(n):  
    if n == 0:  
        print("Launch!")  
    else:  
        print(n)  
        countdown(n - 1)  
n = int(input("Enter the countdown number: "))  
countdown(n)
```

OUTPUT:

A screenshot of an IDLE Shell window titled 'IDLE Shell 3.14.2'. The window shows the execution of the program. It starts with a prompt 'Enter "help" below or click "Help" above for more information.' followed by a restart message. The user enters '10' as the countdown number. The output shows a sequence of numbers from 10 down to 1, followed by 'Launch!'. The code is as follows:

```
>>>  
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1A.py =====  
Enter the countdown number: 10  
10  
9  
8  
7  
6  
5  
4  
3  
2  
1  
Launch!  
>>>
```

EXPLANATION:

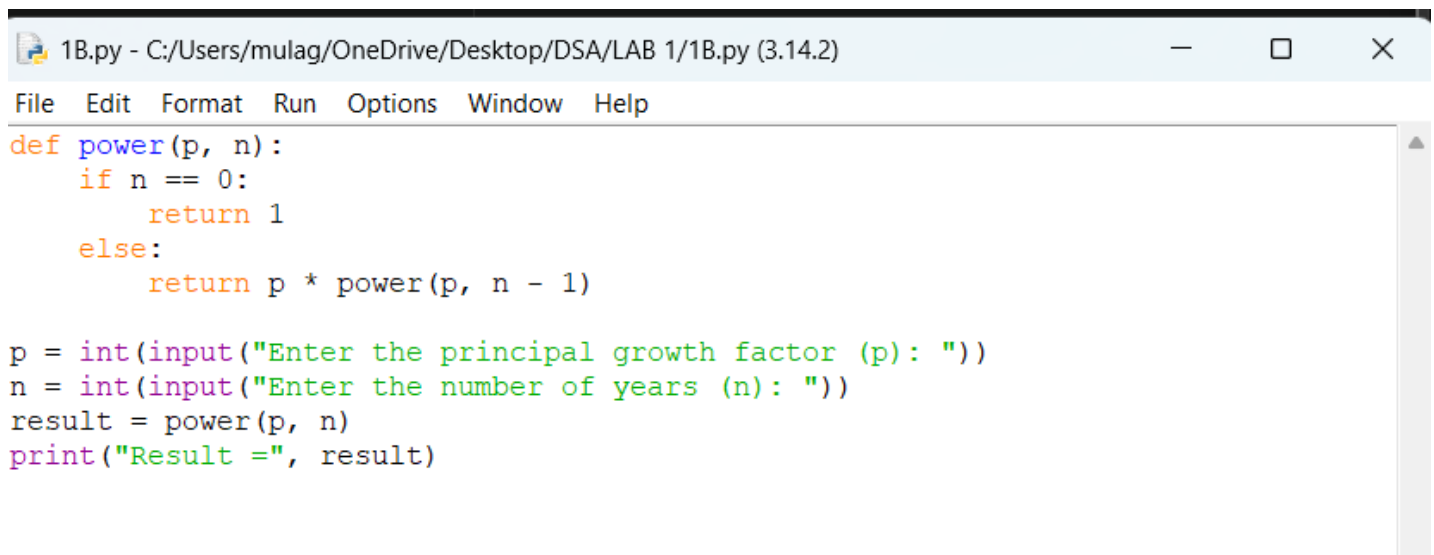
This program displays a countdown from n to 1 and finally prints 'launch'. Although the aim mentions recursion, the given code uses a loop.

ERROR TABLE:

S.NO	ERROR	ERROR RECTIFICATION
1	Indentation Error	Align blocks properly
2	Value Error: invalid literal for int()	Enter a valid integer
3	Name Error: name 'countdown' is not defined	Use the correct function name: countdown().

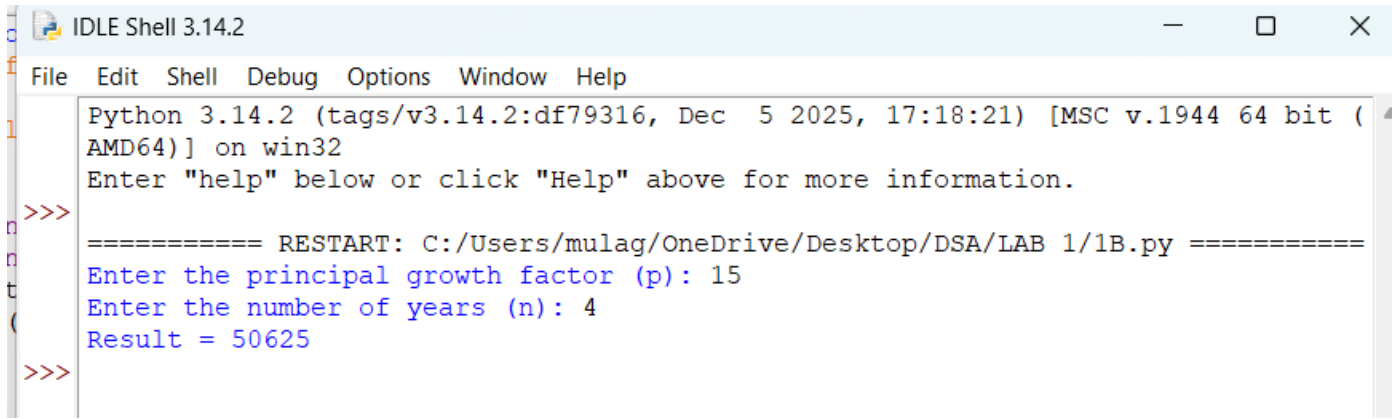
2) AIM: Banks calculate the growth of an investment using compound interest the formula requires calculating powers. Which can be implemented recursively. Write a recursive program to calculate p^n (power), where p =principal growth factor, n = number of years.

PROGRAM:

A screenshot of a Python IDE window titled '1B.py - C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1B.py (3.14.2)'. The window has a menu bar with 'File', 'Edit', 'Format', 'Run', 'Options', 'Window', and 'Help'. The code editor contains the following Python code:

```
def power(p, n):  
    if n == 0:  
        return 1  
    else:  
        return p * power(p, n - 1)  
  
p = int(input("Enter the principal growth factor (p): "))  
n = int(input("Enter the number of years (n): "))  
result = power(p, n)  
print("Result =", result)
```

OUTPUT:



```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1B.py =====
Enter the principal growth factor (p): 15
Enter the number of years (n): 4
Result = 50625
>>>
```

EXPLANATION:

This program recursively calculates p^n by multiplying p until n becomes 0.

ERROR TABLE:

<u>S NO</u>	<u>ERROR</u>	<u>ERROR RECTIFICATION</u>
<u>1</u>	NameError – Function name is spelled incorrectly.	Use the correct function name
<u>2</u>	ValueError – entered letters instead of a number	Enter a valid integer
<u>3</u>	SyntaxError – Missing : after if or else.	Add the colon (:) after if and else statements.

3)AIM:

An HR department stores employee ID's in a list. Develop a recursive program to search for a given employee ID .write a recursive function to search for an employee ID in a list.

PROGRAM:

File Edit Format Run Options Window Help

```

def search_emp(emp_list, key, index):
    if index >= len(emp_list):
        return False
    if emp_list[index] == key:
        return True
    return search_emp(emp_list, key, index + 1)

emp_list = [101, 102, 103, 104, 105]

key = int(input("Enter Employee ID to search: "))

if search_emp(emp_list, key, 0):
    print("Employee ID Found")
else:
    print("Employee ID Not Found")

```

OUTPUT:

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

```

>>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1C.py =====
Enter Employee ID to search: 101
Employee ID Found

>>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1C.py =====
Enter Employee ID to search: 2327
Employee ID Not Found

>>>

```

EXPLANATION: This program searches for an employee ID in a list and returns its index if found.

ERROR TABLE:

S.NO	ERRORS	ERRORS RECTIFICATION
1	Name Error – Function name is typed incorrectly.	Use the correct function name search_emp()
2	Value Error – User enters letters instead of a number.	Enter a valid employee ID (integer), e.g., 103.
3	Syntax Error – Missing : after if statement.	Add : after if and else statements.

4)AIM: In a logistics company the number of possible ways to arrange n different parcels is calculated using the factorial formula($n!$). develop a recursive program to calculate the factorial of a given number

PROGRAM:

```
1D.py - C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1D.py (3.14.2)
File Edit Format Run Options Window Help
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)
n = int(input("Enter a number: "))

if n < 0:
    print("Invalid input! Factorial is not defined for negative numbers.")
else:
    print("Factorial =", factorial(n))
```

OUTPUT:

```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1D.py =====
Enter a number: 5
Factorial = 120
>>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1D.py =====
Enter a number: -1
Invalid input! Factorial is not defined for negative numbers.
>>>
```

EXPLANATION:

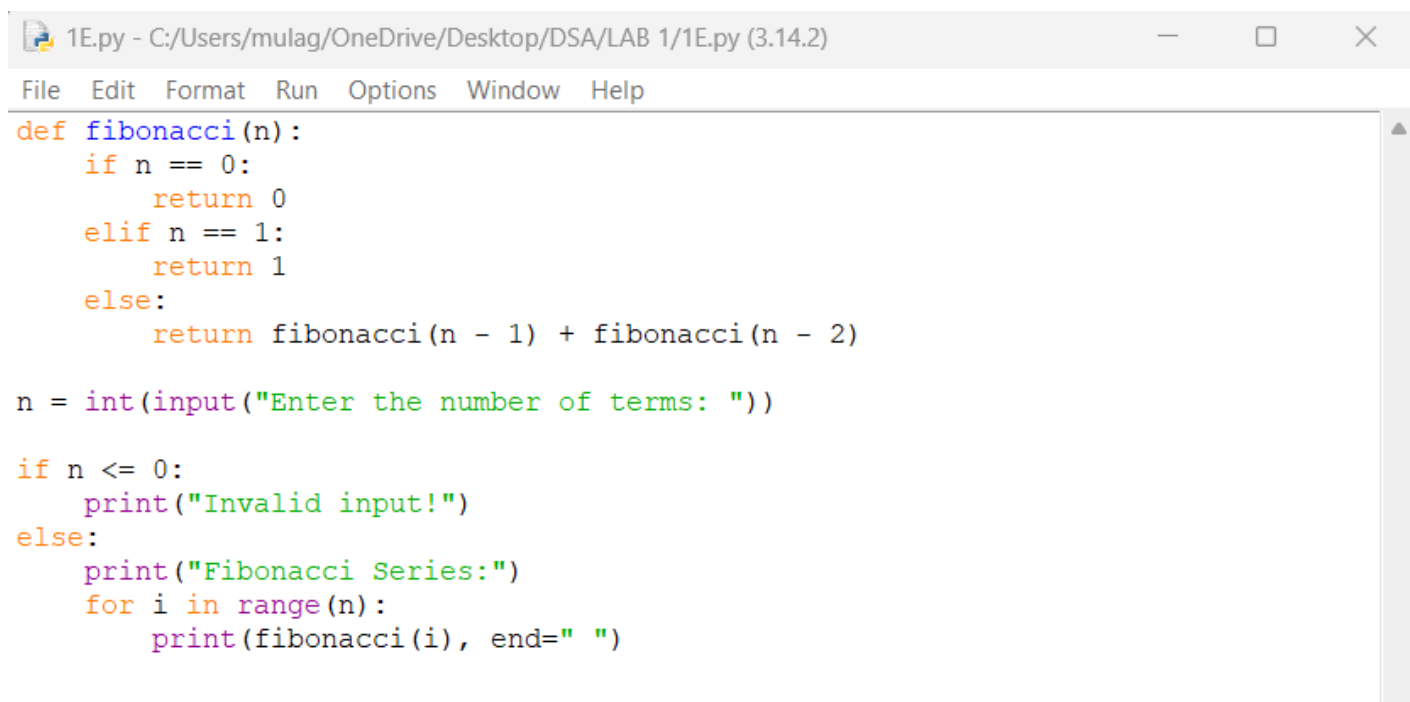
This program calculates the factorial of a number using iteration (not recursion).

ERROR TABLE:

SNO	ERRORS	ERRORS RECTIFICATION
1	Overflow/large input	Give smaller input
2	NameError – Function name is typed incorrectly.	Use the correct function name factorial().

5)AIM: write a recursive python program to print the first n terms of the Fibonacci series.

PROGRAM:

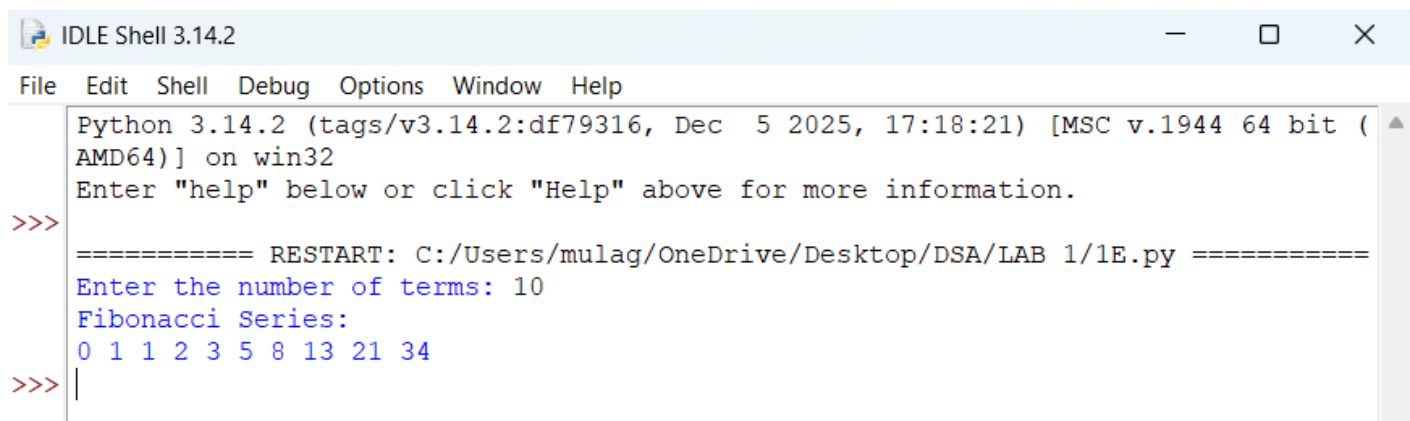


```
1E.py - C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1E.py (3.14.2)
File Edit Format Run Options Window Help
def fibonacci(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

n = int(input("Enter the number of terms: "))

if n <= 0:
    print("Invalid input!")
else:
    print("Fibonacci Series:")
    for i in range(n):
        print(fibonacci(i), end=" ")
```

OUTPUT:



```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 1/1E.py =====
Enter the number of terms: 10
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34
>>> |
```


EXPLANATION:

The program uses recursion to generate Fibonacci numbers. $\text{fib}(0)=0$ and $\text{fib}(1)=1$ are the base cases. For every other term, $\text{fib}(n)=\text{fib}(n-1) + \text{fib}(n-2)$. The loop prints the first n terms.

ERROR TABLE:

SNO	ERRORS	ERRORS RECTIFICATION
1	Recursion Error – Base case is missing or incorrect in the recursive function.	Add the correct base cases: if $n == 0$: return 0 and elif $n == 1$: return 1.
2	Indentation Error – Incorrect indentation after if or else.	Indent the code properly using spaces or tabs.
3	Type Error – Passing a string instead of an integer.	Convert the input using <code>int(input())</code> .

PROGRAM	TIME COMPLEXITY	SPACE COMPLEXITY	REASON
1.countdown to launch	O(n)	O(1)	The program uses a loop that prints from n to 0 once.
2.power calculation	O(n)	O(n)	One recursive call is made for each value of n . The recursion stack stores n calls.
3.employee ID search	O(n)	O(1)	The program performs a linear search through the list.
4.Factorial	O(n)	O(1)	The program calculates factorial using a loop from 1 to n .
5.Fibonacci series	O(2ⁿ)	O(n)	Each function call creates two more recursive calls, leading to exponential time. The recursion stack reaches depth n .

Lab-2

1)Aim: Write a python program to perform linear search on a given input.

Program:

```
2A.py - C:/Users/mulag/OneDrive/Desktop/DSA/LAB 2/2A.py (3.14.2)
File Edit Format Run Options Window Help
def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            return i
    return -1

n=int(input("Enter the no of elements: "))
arr=[]
print("enter the elements:")
for i in range(n):
    arr.append(int(input()))
key = int(input("Enter element to search: "))

result = linear_search(arr, key)

if result != -1:
    print("Element found at position", result)
else:
    print("Element not found")
```

Output:

```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 2/2A.py =====
Enter the no of elements: 6
enter the elements:
3
5
9
8
9
1
Enter element to search: 1
Element found at position 5
>>> |
```

EXPLANATION:

- Takes the number of elements and stores them in a list.
- Takes the target element to be searched.
- linearSearch() checks each element one by one.
- If the target is found, its index is returned.
- If not found, -1 is returned.
- The result is displayed using print().

ERROR TABLE:

S.No	Error Obtained	Rectification
<u>1</u>	Element is not found	Check whether the required element is present in the list
<u>2</u>	Wrong index is displayed	Check the list indexing starting from 0
<u>3</u>	Search stops unexpectedly	Verify the range used in the for loop

2. Aim:

Write a Python program to perform binary search on the given input. Verify by taking sorted array & unsorted array as input.

Program:

2b.py - C:/Users/mulag/OneDrive/Desktop/DSA/LAB 2/2b.py (3.14.2)

File Edit Format Run Options Window Help

```
def binary_search(arr, key):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return -1

n = int(input("Enter number of elements: "))
arr = []
print("Enter", n, "elements:")
for i in range(n):
    arr.append(int(input()))

key = int(input("Enter element to search: "))

result = binary_search(arr, key)

if result != -1:
    print("Element found at position", result + 1)
else:
    print("Element not found")
```

Output:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 2/2b.py =====
Enter number of elements: 5
Enter 5 elements:
10
20
30
40
50
Enter element to search: 40
Element found at position 4
>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/LAB 2/2b.py =====
Enter number of elements: 5
Enter 5 elements:
50
10
30
20
40
Enter element to search: 40
Element found at position 5
>> |
```

EXPLANATION:

- Takes the number of elements and stores them in a list.
- Sorts the list using `a.sort()` because binary search requires sorted data.
- Calculates the middle element and compares it with the target.
- If the target is smaller or larger, searches only the appropriate half.
- Returns the index if found; otherwise returns -1.
- Displays whether the element is found or not.

ERROR TABLE:

S.No	Error Obtained	Error Rectification
1	Wrong element is returned	Make sure the list is sorted before searching
2	Search gives incorrect index	Check the calculation of the middle index
3	Element is not found	Verify the low and high boundaries

Lab-3

1.Aim: Write a Python program to sort the elements using a Bubble sort technique.

Program:

```
3.1.py - C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.1.py (3.14.2)
File Edit Format Run Options Window Help
def bubble_sort(arr):
    n = len(arr)
    for i in range(n - 1):
        for j in range(n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
n = int(input("Enter the number of elements: "))
arr = []
for i in range(n):
    element = int(input(f"Enter element {i + 1}: "))
    arr.append(element)
bubble_sort(arr)
print("Sorted elements:", arr)
```

Output:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.1.py =====
Enter the number of elements: 5
Enter element 1: 40
Enter element 2: 29
Enter element 3: 44
Enter element 4: 35
Enter element 5: 23
Sorted elements: [23, 29, 35, 40, 44]
>>> |
```

Explanation:

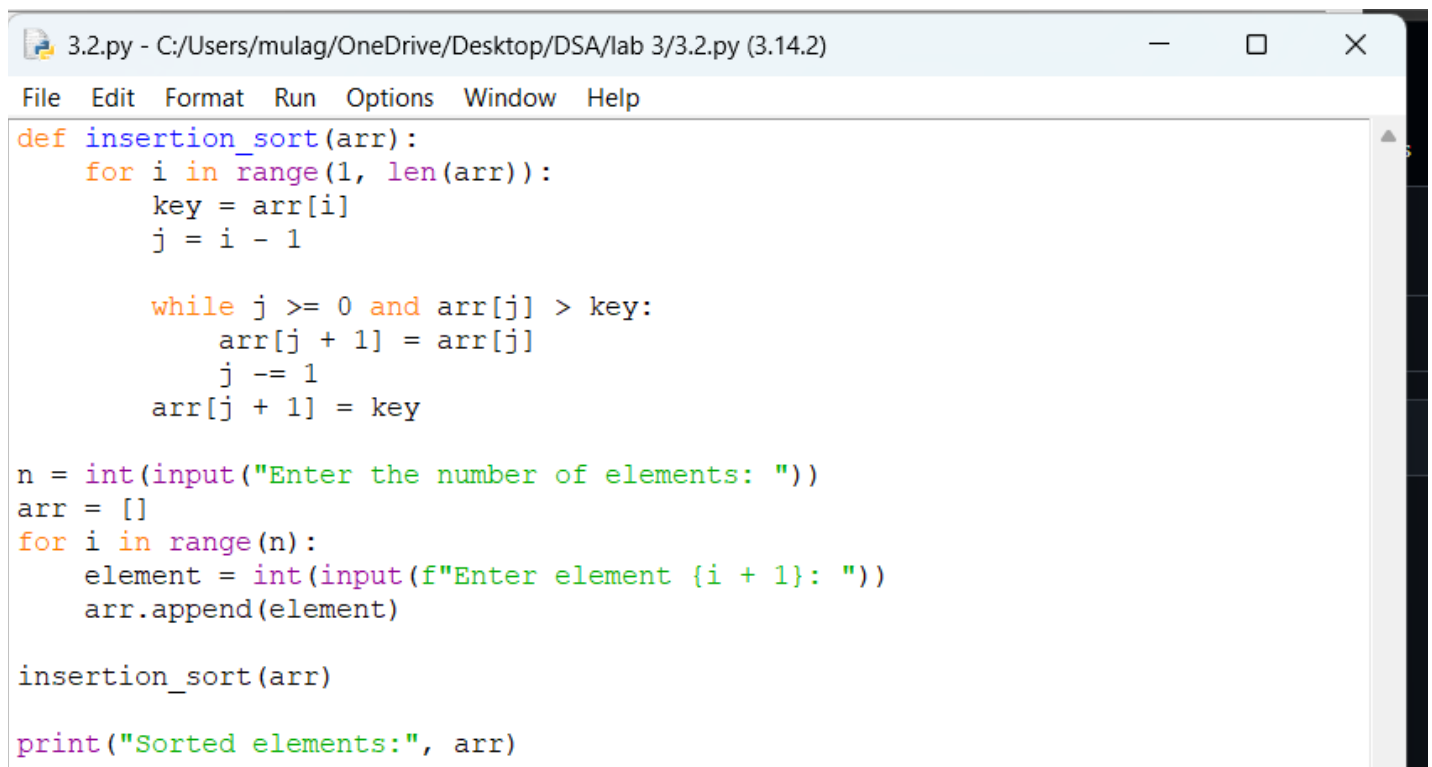
Bubble Sort compares two adjacent elements. If the first element is bigger than the second, it swaps them. This process repeats until all elements are arranged in ascending order.

Error table:

S.No	Error Obtained	Error Rectification
1	NameError because the function name is incorrect	Check that bubble_sort(arr) is written correctly
2	ValueError when entering input	Enter only integer values

2.Aim: Write a Python program to sort the elements using a insertion sort technique.

Program:



```
3.2.py - C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.2.py (3.14.2)
File Edit Format Run Options Window Help
def insertion_sort(arr):
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1

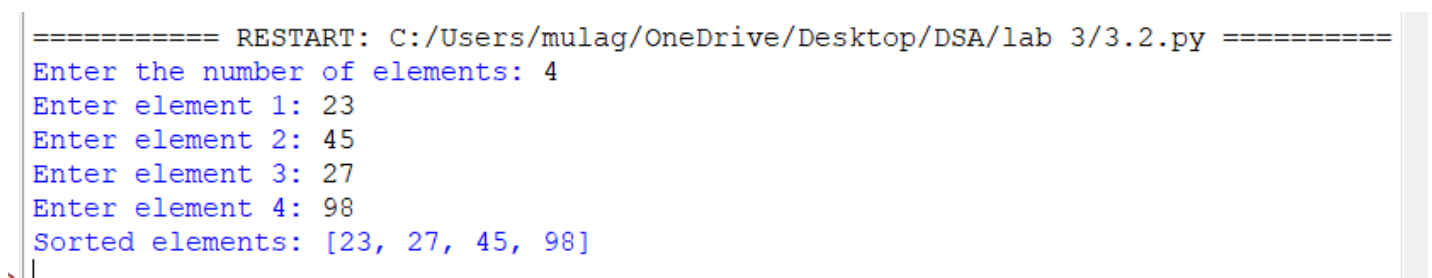
        while j >= 0 and arr[j] > key:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key

n = int(input("Enter the number of elements: "))
arr = []
for i in range(n):
    element = int(input(f"Enter element {i + 1}: "))
    arr.append(element)

insertion_sort(arr)

print("Sorted elements:", arr)
```

Output:



```
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.2.py =====
Enter the number of elements: 4
Enter element 1: 23
Enter element 2: 45
Enter element 3: 27
Enter element 4: 98
Sorted elements: [23, 27, 45, 98]
>|
```


Explanation:

This program sorts the given elements in **ascending order using Insertion Sort**. It takes each element (key) and compares it with the elements before it, shifting larger elements one position to the right. The correct position for the key is then found and the element is inserted there. Finally, the sorted list is displayed using print().

Error table:

<u>S.No</u>	<u>Error Obtained</u>	<u>Error Rectification</u>
1	SyntaxError due to missing colon (:)	Add : after def, for, and while statements
2	IndentationError due to incorrect spacing	Maintain proper indentation inside loops and functions

3.Aim: Write a Python program to sort the elements using a selection sort technique.

Program:

```
3.3.py - C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.3.py (3.14.2)
File Edit Format Run Options Window Help
def selection_sort(arr):
    n = len(arr)
    for i in range(n - 1):
        min_index = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        arr[i], arr[min_index] = arr[min_index], arr[i]
n = int(input("Enter the number of elements: "))
arr = []
for i in range(n):
    element = int(input(f"Enter element {i + 1}: "))
    arr.append(element)
selection_sort(arr)
print("Sorted elements:", arr)
```

Output:

```
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.3.py =====
Enter the number of elements: 4
Enter element 1: 67
Enter element 2: 89
Enter element 3: 34
Enter element 4: 23
Sorted elements: [23, 34, 67, 89]
|
```

Explanation:

This program sorts the given elements in **ascending order using Selection Sort**. It finds the smallest element from the unsorted part of the list and swaps it with the current position. This process repeats until all elements are sorted, and finally the sorted list is displayed.

Error table:

<u>S.No</u>	<u>Error Obtained</u>	<u>Error Rectification</u>
1.	SyntaxError due to missing colon (:)	Add : after def, for, and if statements
2.	IndentationError due to spacing	Maintain proper indentation inside the function and loops
3.	NameError due to wrong variable name	Check the spelling of variables like arr, min_index, i, and j

4.Aim: Write a Python program to sort the elements using a merge sort technique.

Program:

```

def merge_sort(arr):
    if len(arr) > 1:
        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)
        i = 0
        j = 0
        k = 0
        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1
        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1
        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1

n = int(input("Enter the number of elements: "))
arr = []
for i in range(n):
    element = int(input(f"Enter element {i + 1}: "))
    arr.append(element)

merge_sort(arr)

print("Sorted elements:", arr)

```

Output:

```

===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.4.py =====
Enter the number of elements: 6
Enter element 1: 34
Enter element 2: 76
Enter element 3: 89
Enter element 4: 43
Enter element 5: 23
Enter element 6: 27
Sorted elements: [23, 27, 34, 43, 76, 89]

```

Explanation:

This program sorts the given elements in **ascending order using Merge Sort**. It divides the list into two halves repeatedly, sorts each half using recursion, and then merges them in the correct order. Finally, the sorted elements are displayed.

Error table:

S.No	Error Obtained	Error Rectification
1	TypeError while entering invalid input	Enter only integer values
2	NameError because merge_sort is misspelled	Use the same function name merge_sort(arr) everywhere

5.Aim: Write a Python program to sort the elements using a quick sort technique.

Program:

```
3.5.py - C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.5.py (3.14.2)
File Edit Format Run Options Window Help
def quick_sort(arr):
    if len(arr) <= 1:
        return arr

    pivot = arr[-1]

    left = []
    right = []
    for i in arr[:-1]:
        if i < pivot:
            left.append(i)
        else:
            right.append(i)
    return quick_sort(left) + [pivot] + quick_sort(right)
n = int(input("Enter the number of elements: "))
arr = []
for i in range(n):
    element = int(input(f"Enter element {i + 1}: "))
    arr.append(element)

arr = quick_sort(arr)
print("Sorted elements:", arr)
```

Output:

```
===== RESTART: C:/Users/mulag/OneDrive/Desktop/DSA/lab 3/3.5.py =====
Enter the number of elements: 5
Enter element 1: 33
Enter element 2: 44
Enter element 3: 66
Enter element 4: 22
Enter element 5: 11
Sorted elements: [11, 22, 33, 44, 66]
```

Explanation:

This program sorts the given elements in **ascending order using Quick Sort**. It selects the last element as the **pivot**, places smaller elements in the left list and larger or equal elements in the right list, then recursively sorts both lists. Finally, it combines the sorted lists and displays the result.

Error table:

S.No	Error Obtained	Error Rectification
1	TypeError while entering invalid input	Enter only integer values
2	RecursionError due to incorrect recursive calls	Ensure the base condition if <code>len(arr) <= 1</code> : is present

